

APPLIED PHYSICS

Finding independent sets in large-scale graphs with a coherent Ising machine

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Combinatorial optimizers using physical systems, or “Ising machines,” have been intensively studied as a way of computation that is more efficient than current digital computers. Here, we demonstrate that a coherent Ising machine (CIM) based on degenerate optical parametric oscillator (DOPO) pulses can find solutions to large-scale maximum independent set problems, which is an essential optimization problem that is closely related to real-world optimization tasks. We implemented a stable external field for each Ising spin as an interaction between a DOPO pulse and a group of ferromagnetically coupled auxiliary pulses. Consequently, the CIM delivered independent sets for dense graphs with up to 40,000 nodes. We compared the performance of the CIM with that of optimized simulated annealing algorithm implemented on a digital computer and found that the CIM outperformed the central processing unit (CPU) in terms of the time needed to obtain good approximate solutions when the graph size exceeded several thousand nodes.

INTRODUCTION

The progress of digital computers has apparently stagnated, and the search has begun for a more efficient means of computation using (nonsilicon) physical systems. One such means, called the “Ising machine,” solves combinatorial optimization problems on the basis of the Ising model (1). The idea is to convert a given problem into a search for the ground state of the Ising model (2) and solve it using a physical system that simulates the interacting spins in the model. So far, numerous Ising machines have been realized in the form of various physical systems, including superconducting quantum bits (3), single electron devices (4), nano-mechanical devices (5), stochastic nanomagnets (6), complementary metal-oxide semiconductor (CMOS) devices (7, 8), spinwaves (9), surface acoustic waves (10), and superparamagnetic tunnel junctions (11). In addition to such solid-state machines, various photonic Ising machines have been studied (12–24). However, despite the various inventions of machines and proof-of-principle experiments, the advantages of such physical Ising machines over digital computers have not yet been demonstrated for real-world problems.

A coherent Ising machine (CIM) is a photonic Ising machine based on degenerate optical parametric oscillators (DOPO) that are networked by measurement feedback (12–14). It has been shown that a CIM can find solutions to maximum cut problems (13–16), which can be converted into the Ising model without external fields. Our team has demonstrated that a CIM could solve maxcut problems with sizes scaling up to 100,000 nodes, making it the largest Ising machine based on a physical system (16). In addition, this 100,000-spin CIM proved to be more than 1000 times faster than simulated annealing implemented on a central processing unit (CPU) in finding approximate solutions to the maxcut problem of a 100,000-node complete graph. On the other hand, it is known that most real-world applications require both spin-spin interaction and external field terms

when they are formulated in the Ising model (2). In (17), we reported a CIM equipped with external field terms, where the measurement-feedback circuit directly provides the external fields. Using this CIM, a transition between ferromagnetic and anti-ferromagnetic states on a two-dimensional square-lattice forced by an external field was demonstrated. However, in the Ising model converted from the quadratic constraint binary optimization (QUBO) problem, as we state later here, we often observe a large mismatch between the spin-spin interaction terms and the external field terms, which results in poor performance in combinatorial optimization.

Here, we report an application of the CIM to the maximum independent set (MIS) problem, which is one of the problems that require external fields when they are formulated in the Ising model. The MIS problem is an essential graph problem with many real-world applications (see the MIS problems section). Recently, several teams have solved MIS problems on quantum computers based on superconducting qubits (25), Rydberg atoms (26–28), and photons (29). However, the problem sizes that those systems could handle were less than 100, making the experiments at best proof-of-principle demonstrations. Here, we show that the CIM can find good approximate solutions to MIS problems of large-scale dense graphs with sizes scaling up to 40,000. As an alternative to the external fields provided by the measurement-feedback circuit (17), we implemented a stable external field with the “auxiliary spin scheme” that we proposed, where the spin-spin interactions between an Ising spin (for computation) and a group of auxiliary spins mimic the external field. We compared the performance of the CIM with that of optimized simulated annealing (OSA) (30) implemented on a cutting-edge digital computer and found that the CIM exhibited an advantage over OSA in terms of the time-to-target solution (TTT) with approximate solutions as target solutions to the MIS problem when the problem size exceeded several thousand.

RESULTS

MIS problems

An independent set (IS) is a set of vertices in a graph in which no two vertices are adjacent. A MIS is defined as the largest IS for a given graph. It is known that finding the MIS is an NP-hard

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problem, which means that it is unlikely that an efficient algorithm exists for finding the MIS of a graph. The MIS problem is regarded as an essential optimization problem in graph theory and is related to various applications such as matching molecular structures, covering location, and coding theory (31).

The general Hamiltonian for a MIS expressed as a QUBO problem is given by

$$H = -A \sum_i s_i + B \sum_{i,j} A_{ij} s_i s_j \quad (1)$$

where $s_i = \{0, 1\}$, A_{ij} is the adjacent matrix of the graph, and A and B are free parameters used to adjust the weights of the bias terms and the interaction terms. This equation implies that the first term on the right-hand side gives the size of the IS if the second term equals zero (25). This Hamiltonian can be rewritten as follows, using the Ising spin variable $\sigma_i = 2s_i - 1$

$$H = \sum_i h_i \sigma_i + \sum_{i,j} J_{ij} \sigma_i \sigma_j \quad (2)$$

where

$$h_i = -\frac{A}{2} + B \frac{N_i}{2} \quad (3)$$

$$J_{ij} = B \frac{A_{ij}}{4} \quad (4)$$

Here, N_i denotes the number of edges on the i th spin.

As in the case of other combinatorial optimization problems, there are several strong heuristic algorithms to solve MIS problems. For example, the phased local search (PLS) maximum clique algorithm implemented on a CPU can find a best known (BK) solution for a 4000-node dense graph (C4000.5, explained in the Experiments section) in only 285 s (32).

It is generally difficult for an Ising machine, which can be applied to a variety of combinatorial optimization problems, to outperform such a special-purpose solver. Thus, here, we compared the CIM with OSA implemented with 100 threads on a digital computer using MIS as a benchmark problem. Although OSA was originally developed for solving spin-glass problems, which are generally represented by sparse graphs (30), most of the techniques used in it, such as a forward computation of energy change, simplified random number generation, and deterministic traversal, are also effective for solving dense problems. As described in section S1, our OSA exhibits strong performance that is comparable to those of cutting-edge Ising machines based on field programmable gate array (FPGA) [a simulated bifurcation machine (33) and a simulated CIM with chaotic amplitude control (34)] in solving sparse graph problems called Gset (35).

External field by auxiliary spins

Equations 3 and 4 imply that the external field h_i is much larger than the spin-spin interaction J_{ij} , especially with a dense graph. This large difference between the first and the second terms of the Ising Hamiltonian, Eq. 2, often results in a failure to reach low-energy states. The imbalance between h_i and J_{ij} especially degrades the performance of physical Ising machines that tend to have a limited dynamic range in these parameters compared with digital hardware.

To circumvent this problem, we allocate N_{aux} auxiliary spins (or DOPO pulses) in addition to the original Ising spins and modify the

spin-spin interaction matrix as shown in Fig. 1. In so doing, we convert the strong magnetic field term $\sum_i h_i \sigma_i$ in Eq. 2 to many spin-spin interaction terms between the Ising spins and the auxiliary spins, $\sum_{ik} H_{ik} \sigma_i \alpha_k$, where α_k denotes an auxiliary spin. At the same time, we put an all-to-all ferromagnetic interaction among the auxiliary spins with a submatrix F_{kl} . When all of the auxiliary spins are aligned in the same value, we can effectively obtain the magnetic field term as $h_i = \sum_j^{N_{\text{aux}}} H_{ik}$. Consequently, the external field term reduces to $\sum_i (\sum_k H_{ik} \alpha_k) \sigma_i = \alpha \sum_i h_i \sigma_i$, where α denotes the value of the “aligned” auxiliary spins. Thus, the Ising Hamiltonian with the auxiliary spins can be expressed as follows

$$H = - \sum_{k,l} F_{kl} \alpha_k \alpha_l + \sum_{i,k} H_{ik} \sigma_i \alpha_k + \sum_{i,j} J_{ij} \sigma_i \sigma_j \quad (5)$$

Here, F_{kl} determines the strength of the all-to-all ferromagnetic interactions between the auxiliary spins. As a result, the interaction between σ_i and h_i is converted into interactions between σ_i and several α_k with coupling strength H_{ik} , which can be made comparable to J_{ij} by choosing an appropriate number of auxiliary spins N_{aux} .

The imbalance between the external field and the spin-spin interaction terms generally appears in the Ising model converted from a QUBO problem with bias field terms. This means that the auxiliary spin scheme is beneficial not only for the CIM but also for other Ising machines including ones based on digital hardware (33, 34, 36), especially for dense graph problems. In section S2, we show the IS distributions obtained for the C4000.5 graph (see the Experiments section) under the external field and auxiliary spin schemes; the results clearly show the advantage of the auxiliary spin scheme. Thus, we used this scheme not only in the CIM but also in OSA in the following experiments.

Experiments

We used two groups of graphs for evaluating the CIM. The first group included graphs from the second DIMACS Implementation Challenge (37), which has been used as a benchmark instances for MIS or maximum clique problems (32, 38, 39). We especially focused on finding MISs on large-scale graphs with 1000 or more nodes. The largest instance (C4000.5) had 4000 nodes with an edge density of 50%. The second group consisted of randomly generated graphs with a 50% edge density and problem sizes ranging from 500 to as many as 40,000.

The MISs of highly dense graphs tend to be much smaller than their problem sizes. For example, the BK score of C4000.5 is 18, which means that only 18 spins take a value of 1 and all other spins

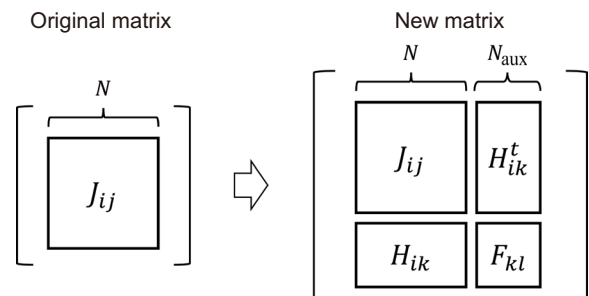


Fig. 1. Matrix for the auxiliary spin scheme. J_{ij} corresponds to an $N \times N$ spin-spin interaction matrix for a given graph (N : graph size). H_{ik} is an $N \times N_{\text{aux}}$ matrix that connects the Ising spins and auxiliary spins, where N_{aux} denotes the number of auxiliary spins. F_{kl} is an $N_{\text{aux}} \times N_{\text{aux}}$ matrix that enables ferromagnetic all-to-all connections among N_{aux} auxiliary spins.

take a value of -1 . Since the balanced homodyne measurement system in the CIM is AC-coupled, we observed huge fluctuations of the baseline level during the search of IS. Since we converted the analog amplitudes of the final DOPO pulses to a “spin value” by thresholding at 0 V, the random fluctuation of the baseline caused errors in spin readout, resulting in a failure of the computation. To circumvent this problem and stabilize the baseline of the measurement, we multiplied J_{ij} of the given graph with a random spin-flip matrix so that we can suppress the DC component in the measurement result at the homodyne detector and thus stabilize the baseline level.

When the graph is small, we can put the same graph repeatedly in 49,744 signal pulses of the CIM so that we can simultaneously compute the MIS of the same graph many times in each CIM computation. Hereafter, we denote the number of embeddings of a problem graph as N_{emb} , which is given as the quotient of 49,744 and the graph size including auxiliary spins for external fields. In one run, the CIM undertakes 32 computations. For each graph, we ran the CIM 100 times, which means that we obtained $3200 \times N_{\text{emb}}$ results for each graph. In the experiments, the value of N_{emb} ranged from 1 (for 40,000-node graphs) to 90 (for 500-node graphs).

As a metric to compare the performances of the CIM and OSA, we used TTT, which is defined as

$$T_{\text{tts}} = \frac{T_{\text{ann}}}{N_{\text{emb}}} \frac{\log(1-p_t)}{\log(1-p)} \quad (6)$$

where T_{ann} , p_t , and p denote the time for one computation (annealing time), the target success probability, and the success probability to obtain the target score, respectively. In our experiment, we set $p_t = 0.99$. T_{ann} of OSA is the time of a parallel simulation using 100 threads on a CPU. In many cases, the CIM (and sometimes also OSA on a CPU) fails to reach the BK IS scores obtained by special-purpose solvers. Because of that, we used approximate solutions as the target scores for TTT.

We experimentally investigated how the performance changes depending on the number of auxiliary spins N_{aux} . We searched the ISs of the C4000.5 graph in the DIMACS graph set while varying N_{aux} . Figure 2 shows the result, which is the probability of obtaining an IS that is greater than or equal to 14 as a function of N_{aux} . The blue and orange dots denote the raw data and filtered data where the Ising energies of the monitor graph were less than or equal to -40 . When $N_{\text{aux}} \leq 100$, the probability steeply increased and peaked at $N_{\text{aux}} = 200$. Then, the probability decreased as N_{aux} became larger. We performed similar measurements on several other graphs with various sizes and found that the optimum N_{aux} was around 5% of the problem size. In the following experiments, we set N_{aux} to 5% of the problem size. Note that we could not find ISs larger than or equal to 14 for the C4000.5 graph with the external field term directly provided by the measurement-feedback circuit (17). Thus, we concluded that the auxiliary spin scheme improves the probability of reaching a good solution. For OSA, we set N_{aux} at 10% of the problem size by using a similar procedure.

The reason that the probability to obtain good ISs shown in Fig. 2 forms peak structures can be intuitively explained as follows. By increasing the number of auxiliary spins, we can effectively increase the magnetic field, which results in better performance in solving problems. On the other hand, by increasing N_{aux} , we enlarge the size of the Ising problem, which results in degradation of the success probability and TTT. Because of the competition between these two mechanisms, the success probability as a function of N_{aux} exhibits a peak structure.

We searched the ISs of the DIMACS graphs whose sizes were greater than or equal to 1000 by using the CIM and OSA, both of which were based on the Ising model with the auxiliary spin scheme. In OSA, we implemented automatic rescaling of the J_{ij} weights. The rescaling factor r is $r = \sum_{ij} |J_{ij}|^2 / N$, where N denotes the graph size. By dividing J_{ij} by r , we can roughly optimize the interaction strength to obtain low-energy states depending on the graph size. To obtain the TTTs, we ran OSA 100 times for each graph with five different step numbers, i.e., $N_{\text{stp}} = \{100, 1000, 10000, 100000, 1000000\}$, and took the run with the best TTT for obtaining an IS that is greater than or equal to the target score. Here, the step number N_{stp} is the number of repeated annealing processes, where an annealing process corresponds to N -time spin-flip trials. The MISs obtained by the CIM, OSA, and the best-known MIS are summarized in Table 1, and the detailed data are shown in section S3. OSA generally outperformed the CIM in obtaining a larger IS, which is as expected because the annealing time can be arbitrarily extended on a digital computer. On the other hand, it is difficult to keep the dynamics of the DOPO amplitude evolution for a long time while at the same time avoiding saturation of the amplitude. The gap in MIS between the CIM and OSA (or BK) became wider when the edge density of the given graph was small, whereas for graphs with high edge densities (0.9, with C2000.9), the CIM could find the best-known MIS. Next, we compared the TTTs of the CIM and OSA with the best scores obtained by the CIM as the target scores. The results, shown in Fig. 3, indicate that OSA was faster than the CIM in most graph instances, but that the

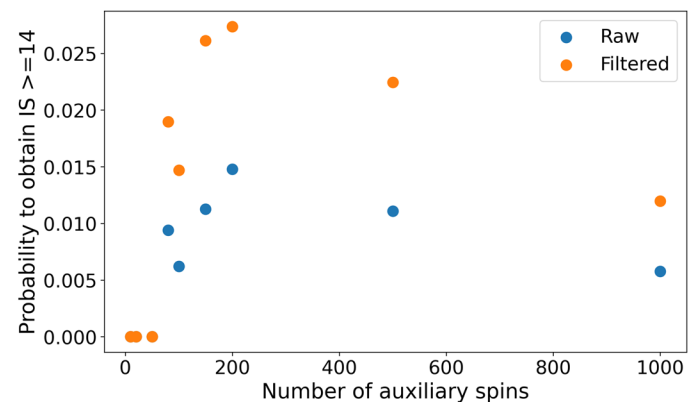


Fig. 2. Probability of obtaining IS larger than or equal to 14 for C4000.5 graph as a function of the number of auxiliary spins. The “filtered” data (orange dots) denote the probability calculated by picking the data for which the Ising energies of the monitor graphs were less than or equal to -40 .

CIM slightly outperformed OSA on the largest graph in the set (C4000.5).

The TTTs for the DIMACS graphs imply that the CIM may have an advantage in finding good approximate solutions to dense graph problems with sizes of several thousand or larger. To investigate the performance of the CIM and OSA on even larger graphs, we used a random-edge-selecting algorithm to generate random graphs with 50% edge densities and sizes ranging from 500 to 40,000. We made five instances for each graph size. We ran the CIM and OSA for each instance and obtained average TTT values for each graph size. Here, the

Table 1. Obtained MIS for DIMACS graphs.								
	C1000.9	DSJC1000_5	p_hat15000-1	p_hat1500-2	p_hat1500-3	C2000.5	C2000.9	C4000.5
CIM	6	14	77	58	11	15	6	16
OSA	6	15	87	62	11	16	6	17
Best known (BK)	6	15	87	65	–	17	6	18
Ref. of BK	(32)	(38, 39)	(32)	(32, 38, 39)	–	(32)	(32)	(32, 38)
Edge density	0.9	0.5	0.13	0.25	0.38	0.5	0.9	0.5

best score obtained with the CIM was used as the target score. We ran OSA 100 times for each graph with $N_{\text{stp}} = \{100, 1000, 10000, 10000\}$ and calculated the TTTs to obtain the target solutions. The results are shown in Fig. 4A, and the detailed data are provided in section S4. Consequently, the CIM outperformed OSA on graphs larger than about 5000. Figure 4B shows the same data from a different perspective: an instance-by-instance comparison of TTTs. It is clear that the CIM is faster for all the instances with sizes of 10,000 or larger. These observations suggest that, as the graph size becomes larger, it becomes increasingly difficult for OSA to obtain approximate solutions that are as good as the CIM can deliver.

To investigate the performance of the CIM with that of OSA as a reference, we plotted the TTT with 0.9 of the best scores obtained by OSA as target scores in Fig. 4C and an instance-by-instance comparison in Fig. 4D. Here too, we confirmed that the CIM tends to outperform OSA as the graph size becomes larger.

DISCUSSION

In Fig. 4A, the TTT of the CIM does not seem to depend on the graph size. This is because the best scores of the CIM are used here as the target. In this case, for example, the TTTs for smaller graphs are not necessarily smaller than those of larger graphs, because the Ising energy obtained by the CIM in a fixed annealing time may get closer to the ground states as the graph size becomes smaller. Consequently, the accuracy of the solutions obtained by the CIM is likely to degrade as the graph size becomes larger. Even with the target scores that “deteriorate” as the graph size becomes larger, it seems that the TTT of OSA steadily increases.

Here, we chose OSA as a counterpart of the CIM, based on its strong performance compared with cutting-edge digital Ising machines as described in section S1. An important future work will be a comprehensive benchmarking against various schemes to clarify the advantages of the CIM over existing techniques implemented on digital hardware. The schemes to be compared include Ising model solvers based on parallel tempering (40, 41) and PLS-based algorithms specialized in maximum clique (32).

While the use of multiplication of a random matrix is necessary in the current CIM to circumvent the baseline fluctuation caused by the AC-coupled measurement setup, the multiplication of such a matrix may influence the optimization process in the computation by the CIM, resulting in a change in the success probability. An important future work will thus be a detailed investigation of the success probability dependence on instances of random matrices.

The obtained results confirm that the CIM can quickly find good approximate solutions to problems with large sizes and dense structures. A natural question is: What is an application where this characteristic is most effectively used? Such an application would require a fast and frequent optimization of large-scale networks, but not necessarily exact solutions. Candidate applications include optimization of a next-generation wireless network (42) and accurate object detection (43).

MATERIALS AND METHODS
Experimental setup and procedure

Figure 5 shows the configuration of the CIM used in the experiments. The CIM consisted of a phase-sensitive amplifier (PSA), two

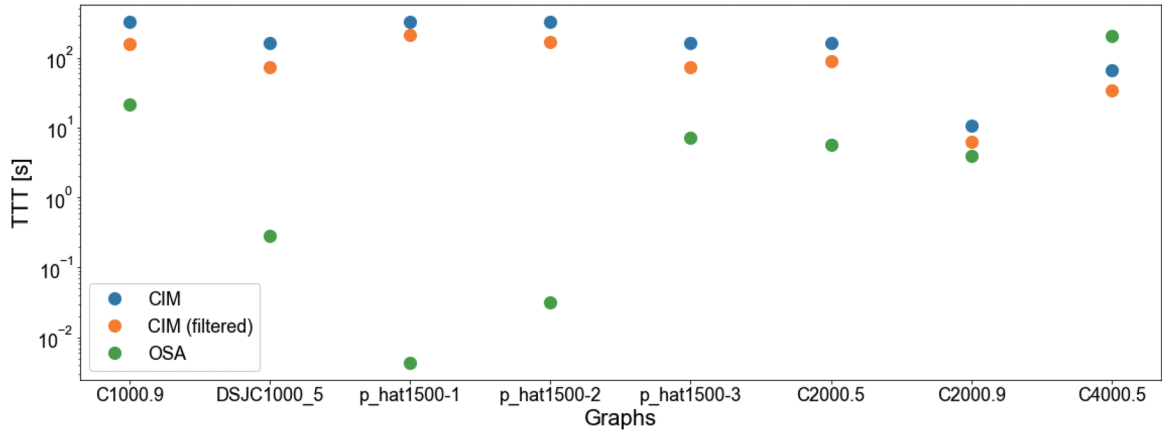


Fig. 3. Time-to-target solution (TTT) with the best scores obtained by the coherent Ising machine (CIM) as target scores (DIMACS graphs). The orange dots (“filtered data”) denote hypothetical TTTs obtained by picking the data for which the Ising energies of the monitor graphs were less than or equal to −40.

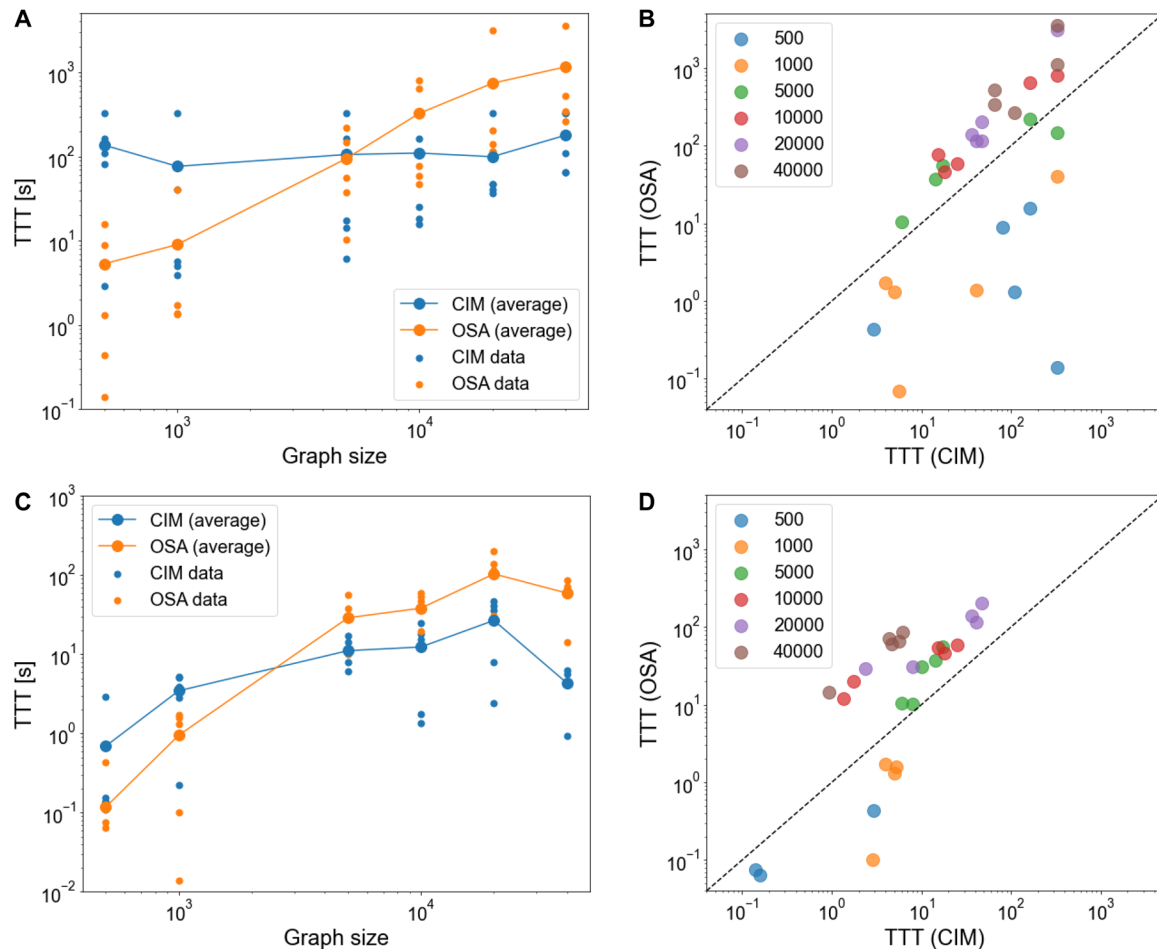


Fig. 4. Time-to-target solution (TTT) obtained with random graphs. (A) Graph-size dependence of TTT with the best scores obtained by the coherent Ising machine (CIM) as target scores. Five instances were generated for each graph size. The larger dots show the average TTT of five instances. (B) Instance-by-instance comparison of TTT obtained by the CIM and optimized simulated annealing (OSA) (target score: best scores of the CIM). (C) Graph-size dependence of TTT with 0.9 of the best scores obtained by OSA as target scores. (D) Instance-by-instance comparison of TTT obtained by the CIM and OSA (target score: 0.9 of OSA best scores).

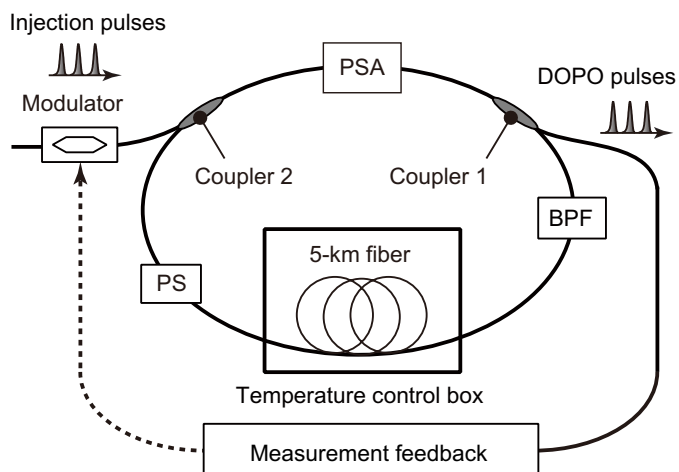


Fig. 5. Configuration of coherent Ising machine (CIM). The measurement-feedback part consists of a balanced homodyne detector for a quadrature measurement of degenerate optical parametric oscillator (DOPO) pulses, an analog-digital converter, a matrix multiplication system based on an FPGA, and a digital-analog converter. PSA, phase-sensitive amplifier; BPF, band-pass filter; PS, phase shifter.

9:1 fiber couplers, an optical band-pass filter, a 5-km fiber, a phase shifter, and a measurement-feedback circuit that includes a balanced homodyne detector, an analog-digital converter, an FPGA system, and a digital-to-analog converter. These optical parts are the same as those in the CIM reported in (16) where 100,512 DOPO pulses were accommodated to simulate the Ising spins with all-to-all coupling with a resolution of two bits [for details, see figure S1 in (16)]. For the present experiments, we reprogrammed the logic of the FPGA for measurement feedback so that the coupling resolution was increased to eight bits (integer between -127 and 127) at the expense of reducing the number of all-to-all coupled pulses to 50,256.

The total number of the DOPO pulses (and blank slots explained below) in the fiber cavity is 123,572, and the temporal spacing between pulses is 200 ps, which means that the round-trip time of the cavity t_r is $\sim 24.7 \mu\text{s}$. The odd-number pulses in the first 100,512 pulses are used as “signal pulses”, on which we put spin-spin interactions of a given Ising model instance by using measurement feedback (13, 14, 16), while the even-number pulses are not used for computation (see Fig. 6 for details). The amplitudes of the signal pulses are periodically controlled using time-varying pump pulses whose temporal schedule is shown in Fig. 7, while the even-number

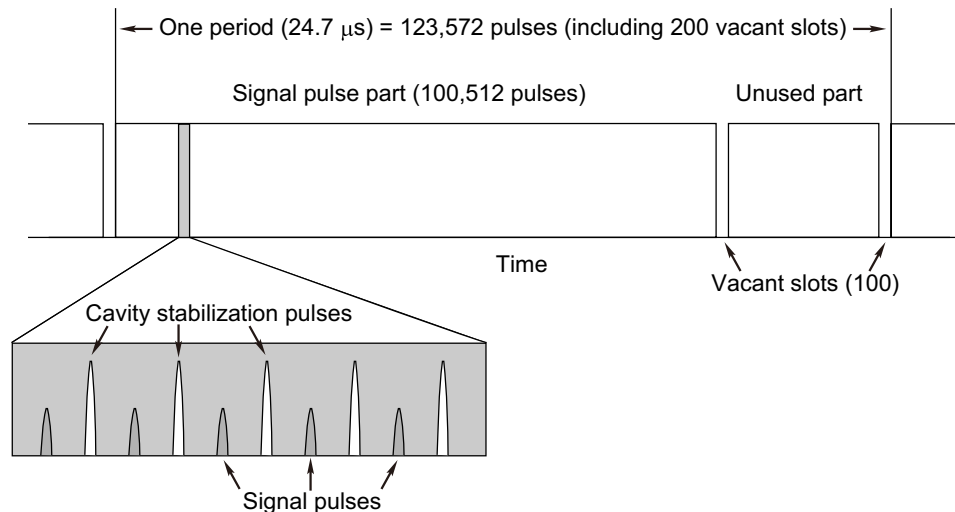


Fig. 6. Structure of time-multiplexed degenerate optical parametric oscillator (DOPO) pulses. One period corresponds to the round-trip time of the fiber cavity shown in Fig. 5. The signal pulse part consists of 100,512 DOPO pulses, where the odd-number pulses are used for computation (signal pulses) and the even-number ones are simply kept oscillating to improve the stability of the cavity (cavity stabilization pulses). This part is followed by 100 blank slots, and then 22,860 unused pulses are generated, which also helps to stabilize the cavity. Finally, 100 blank slots are placed to recognize the position of the start of the next signal pulse part.

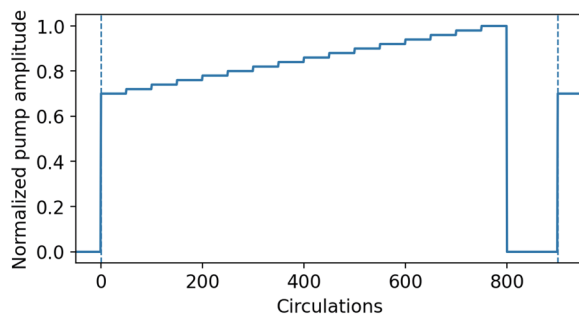


Fig. 7. Pump schedule for the signal pulses. The vertical axis shows the amplitude normalized by the pump amplitude for an uncoupled degenerate optical parametric oscillator (DOPO) pulse at the oscillation threshold. The normalized pump amplitude is increased from 0.7 to 1.0 during 800 circulations in the cavity. Then, we turn off the pump for 100 circulations to eliminate the DOPO light, before starting the next computation. The final spin readout is done at the 750th circulation.

pulses are kept oscillating. By placing stably oscillating unused pulses between pulses that are periodically turned on and off, the stability of the cavity is improved because of the reduced transient response of the erbium-doped fiber amplifier (EDFA) for amplifying the pump pulses. Then, after 100 blank slots (to recognize the positions of these pulses), 22,860 unused pulses are generated, and these pulses are separated by another 100 blank slots.

The first 512 pulses of the 50,256 signal pulses are fixed header pulses. Using 32 pulses in the header, we put a 32-node bipartite graph (monitor graph) through measurement feedback so that we can run the bipartite graph problem simultaneously with a computation on the CIM. The Ising energies of the monitor graph ranges from -88 to 88 , which means that an observation of a positive Ising energy implies that the optical phases of the injection pulses are inverted. Thus, by checking the Ising energy of the monitor graph, we can judge if the CIM is in reasonable condition. The remaining 49,744 pulses are used for solving the MIS problems.

The temporal schedule of the pump pulse amplitude for the signal pulses is shown in Fig. 7, where the vertical axis shows the amplitude normalized by the pump amplitude for an uncoupled DOPO pulse at the oscillation threshold. We increased the normalized pump amplitude from 0.7 to 1.0 during 800 circulations in the cavity. Then, we turned off the pump for 100 circulations to eliminate the DOPO light, before starting the next computation. Thus, one CIM computation took 900 circulations, i.e., $T_{\text{ann}} = 900 \times t_r = 22.23$ ms. The final spin states are read out at the 750th circulation.

Supplementary Materials

This PDF file includes:

Sections S1 to S5
Figs. S1 to S3
Tables S1 to S7

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Finding independent sets in large-scale graphs with a coherent Ising machine

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